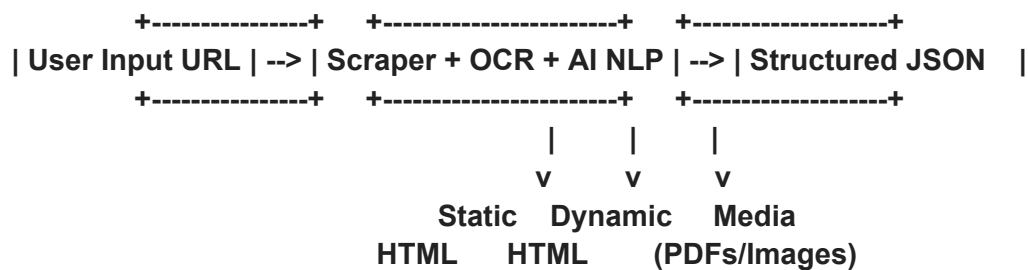


Zomato Gen AI Internship Assignment solution

By- Harsh Kumar

Zomato Scraper – Technical Documentation

System Architecture Explanation



- Input: Zomato restaurant base URL
- Core Components:
 - HTML Scraper: For info, menu, and reviews pages
 - Dynamic Content Renderer: Selenium-based browser emulator to capture JS content
 - Media Extractor: OCR on images and PDFs using **pytesseract** and **pdfplumber**
 - AI Formatter: Uses Cohere's LLM to convert raw text to structured JSON

Implementation Details and Design Decisions

Tech Stack:

- Libraries: `requests`, `BeautifulSoup`, `selenium`, `pytesseract`, `pdfplumber`, `cohere`
- Headless Browser: Chromium via `selenium` to capture JS-rendered content
- AI Model: Cohere's `command-r-plus` to parse and format extracted data
- OCR Handling: Fallback to image processing if text extraction fails from PDFs

Design Decisions:

- Multi-source data gathering: Combining static HTML, JavaScript content, and media ensures data completeness.
- Cohere for parsing: Faster than hand-coding complex parsers for unstructured Zomato data.
- Headless execution: Enables cloud execution (e.g., Google Colab) without UI dependency.

Challenges Faced and Solutions Implemented

Challenge	Solution
Menus were not scrapeable via	ium to render JS in headless Chrome
on low-quality menu images	PDF fallback using pdfplumber and pdf2image
les lacked extractable text	as fallback using pytesseract on each page
d review & menu data	rompt engineering with Cohere to format into
d parsing errors	l try/except blocks and page wait times

Future Improvement Opportunities

- **Rate Limiting & Retry Logic:** Add exponential backoff for failed requests.
 - **Language Support:** Extend OCR to support regional Indian languages in menus.
 - **Scraping Optimization:** Use multi-threading or async scraping for faster execution.
 - **Validation Layer:** Add schema validation (e.g., `pydantic`) to check AI JSON output.
 - **Local AI Model Option:** Use open-source LLMs like LLaMA2 or GPT4All for offline parsing.
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